

COLLEGE OF ENGINEERING THALASSERY



FINAL PROJECT REPORT ON

LoRA BASED V2X COMMUNICATION SYSTEM

Submitted partial fulfillment of the requirement for the award of Degree of Bachelor of Technology in Electronics and Communication Engineering of APJ Abdul Kalam Technological University

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CERTIFICATE

This is to certify that the final-project entitled “LoRA BASED V2X COMMUNICATION SYSTEM” done by Avinash E, Adhish E, Abhinav A, Ashwin k k during the year 2026 in the partial fulfillment for the award for the degree of the Bachelor of Technology in Electronics and Communication Engineering of the APJ Abdul Kalam Technological University.

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DECLARATION

We hereby declare that the mini project report “**LoRA BASED V2X COMMUNICATION SYSTEM**”, submitted for the fulfilment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Assistant Prof. Musthafa T A in Electronics and Communication branch College of Engineering Thalassery. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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OBJECTIVE

To design and develop a LoRa-based V2X communication system that enables vehicles to exchange safety-critical information such as inter-vehicle distance, traffic signal status, pedestrian crossing alerts, and emergency vehicle approach warning.

ABSTRACT

The rapid growth of vehicular networks and intelligent transportation systems has led to the need for reliable and low-cost Vehicle-to-Everything (V2X) communication solutions that enhance road safety and traffic efficiency. This project presents the design and development of a LoRa-based V2X communication system that enables information exchange among vehicles, roadside infrastructure, and pedestrians. The proposed system integrates Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), and Vehicle-to-Pedestrian (V2P) communication into a unified framework using the long-range, low-power LoRa technology. In the V2V module, each vehicle is equipped with an On-Board Unit (OBU) that measures its distance from nearby vehicles using sensors and transmits this data via LoRa to neighboring vehicles. This ensures real-time collision warning when two vehicles approach within a predefined safety range. The V2I communication utilizes a Roadside Unit (RSU) that interacts with the vehicles to provide updates on traffic light status, road conditions, and speed limits, allowing drivers to anticipate upcoming traffic situations. In the V2P module, pedestrian crossings or mobile nodes broadcast signals to alert nearby vehicles of pedestrian movement, reducing the risk of accidents in high-traffic zones. Additionally, an emergency vehicle alert system enables ambulances or priority vehicles to send warnings to nearby OBUs, facilitating timely lane clearance and reduced response time.

This project demonstrates that LoRa-based V2X communication provides a cost-effective alternative to conventional DSRC and cellular V2X systems. By enabling continuous awareness among vehicles, infrastructure, and pedestrians, the proposed system significantly enhances road safety and contributes to the realization of smarter, more connected transportation networks.

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CHAPTER 1

INTRODUCTION

The evolution of modern transportation has brought tremendous convenience, but it has also introduced serious challenges related to road safety, traffic congestion, and emergency response delays. With the growing number of vehicles on the road, traditional safety mechanisms that rely solely on human awareness and visual cues are becoming insufficient. This has led to the emergence of Vehicle-to-Everything (V2X) communication, an advanced concept in intelligent transportation systems where vehicles continuously exchange information with other vehicles, roadside infrastructure, and even pedestrians to create a safer and more coordinated traffic environment.

At its core, V2X aims to allow vehicles to “see” beyond the line of sight by sharing vital information such as position, speed, direction, and road conditions. The two most fundamental components of V2X are Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication. V2V communication allows vehicles to detect and avoid potential collisions, while V2I provides updates on traffic lights, road hazards, and signal timings. Extending this further, Vehicle-to-Pedestrian (V2P) communication ensures that vulnerable road users can also participate in the connected ecosystem. Together, these technologies form the foundation of cooperative and automated transportation systems. Conventional V2X communication technologies such as Dedicated Short Range Communication (DSRC) and Cellular V2X (C-V2X) offer high-speed data transfer but come with drawbacks such as high cost, network dependency, and limited accessibility in rural or low-infrastructure areas. To overcome these limitations, LoRa (Long Range Radio) technology presents a practical alternative. LoRa operates on unlicensed frequency bands and supports long-distance, low-power, and low-data-rate communication, making it ideal for exchanging simple safety-related messages over several kilometers with minimal infrastructure. In this project, a LoRa-based V2X communication system is developed to demonstrate how low-cost wireless technology can enhance vehicular awareness and safety. Each vehicle is equipped with a LoRa transceiver module and a distance measurement unit to estimate its distance from surrounding vehicles. The information is processed by a microcontroller and transmitted to nearby vehicles through LoRa, enabling real-time collision warnings and safety alerts. A Roadside Unit (RSU) is also implemented to relay traffic signal status, pedestrian crossing information, and emergency vehicle alerts to vehicles in its range.

The motivation behind this system lies in developing a reliable, energy-efficient, and affordable communication framework suitable for both urban and rural transportation environments. By leveraging LoRa's long-range capability and simple network setup, the proposed V2X system bridges the gap between advanced vehicular networks and practical, deployable solutions for developing regions. It demonstrates how smart connectivity and cooperative awareness can drastically improve traffic management, road safety, and emergency response, paving the way for future autonomous and intelligent mobility systems.

CHAPTER 2

LITERATURE REVIEW

1. The paper “**A Study of LoRa Low Power and Wide Area Network Technology**” by Umer Noreen, Ahcene Bounceur, and Laurent Clavier provides a comprehensive study of LoRa , a Low Power Wide Area Network protocol designed for Internet of Things applications. The authors begin by situating LoRa within the broader context of wireless sensor networks , which require energy-efficient, low-cost and long-range communication for applications in smart cities, environmental monitoring, and industrial automation. LoRa uses chirp spread spectrum modulation, which provides excellent receiver sensitivity and long communication range at low transmission power. These characteristics align well with V2X scenarios such as vehicle-to-vehicle and vehicle-to-infrastructure communication, especially in low-infrastructure environments where traditional cellular V2X or 5G systems may be unavailable. The paper’s discussion on spreading ,bandwidth and coding rate highlights how parameter selection affects range and reliability critical factors in vehicle communication networks where both message delivery and timing precision are vital. For V2X, Class C operation could enable near-continuous listening for roadside units or gateways, while Class A devices could support low-power sensing units embedded in vehicles. Although LoRa’s low data rate limits its suitability for high-throughput safety messages, its long-range and low-power operation make it promising for non-critical V2X tasks such as traffic monitoring, environmental sensing, vehicle tracking, and status reporting.

2. The paper “**Intelligent Traffic Management Systems**” by Hesham A. Sakr, PLVAR Team, and Magda I. El-Afifi , explores how rapid urbanization and rising vehicle ownership have outpaced road infrastructure development, leading to severe traffic congestion, accidents, pollution, and inefficiencies. Traditional traffic control methods are no longer adequate, so the study reviews the growing adoption of ITMS powered by IoT, AI, Machine learning and sensor networks. The review categorizes ITMS solutions into traffic flow management, congestion prevention, and trip time prediction. Key benefits include predictive insights, enhanced safety, through vehicle-to-infrastructure communication and adaptive signaling.

The paper also outlines the core components of ITMS: smart road sensors (RFID, air-quality, and temperature), connected CCTV, GIS, cloud-based control platforms, and AI-driven analytics such as computer vision and reinforcement learning. Future directions highlight the need for integrated ecosystems, combining multimodal data from vehicles, infrastructure, and personal devices to enable real-time traffic monitoring, predictive planning, and automated parking. The paper emphasizes that while technologies such as smart traffic lights, video analytics, and autonomous vehicle integration are promising, success depends on collaboration between governments, private companies, and urban planners. Ultimately, ITMS represent a key enabler for safer, greener, and more efficient smart cities, but require sustained investment and innovation to address the unpredictability of urban mobility.

3. The paper “**Vehicle to Vehicle Communication with LoRa network**” by Filipe Cortes, Rafael Carvalho and Luis Miguel Pires focus on vehicle-to-vehicle communication using LoRa technology as a way to improve road safety. Traditional V2V systems usually rely on Wi-Fi or cellular networks, but these can be costly, energy-intensive, and less reliable over long distances. LoRa, being a low-power wide-area network, offers long range, low energy consumption, and robustness against interference, which makes it an attractive alternative. The work builds on the broader concept of the Internet of Vehicles, where cars exchange information such as speed, position, or alerts to prevent accidents. Earlier studies are particularly important here. Torres et al. in 2021 experimentally demonstrated that LoRa can indeed support V2V communication, but its effectiveness depends on the spreading factor. Building on this foundation, the authors of this paper designed an experimental prototype using two mobile robots to simulate vehicles. The lead robot was equipped with multiple sensors, including ultrasonic, temperature, humidity, gas, and pulse sensors. It transmitted this information via LoRa to the follower robot, which then adjusted its speed and behavior, for example slowing down when conditions became unsafe. LoRa’s performance is strongly dependent on spreading factor and environmental conditions. Shorter distances worked best with lower spreading factors, while longer distances required higher ones, though at the cost of data rate. Overall, the study shows that LoRa is a feasible option for vehicle-to-vehicle communication and could play a role in building safer, smarter transportation systems.

4. The paper “**V2I-Based Cooperation Between Traffic Signal and Approaching Automated Vehicles**” by Biao Xu, Xuegang (Jeff) Ban, Yougang Bian, Jianqiang Wang, and Keqiang Li explores the use of Vehicle-to-Infrastructure communication to enhance coordination between automated vehicles and traffic signal control systems. The study addresses a fundamental challenge in intelligent transportation how to optimize traffic flow and energy efficiency when vehicles and infrastructure can exchange information in real time. The authors propose a cooperative control framework where approaching vehicles communicate with traffic lights through a V2I channel, allowing both entities to adjust their behaviors dynamically. Vehicles receive Signal Phase and Timing (SPaT) data from the traffic controller, enabling them to optimize acceleration and deceleration profiles to minimize idling and harsh braking. Similarly, the infrastructure can adapt signal timing based on the predicted arrival times and speeds of connected vehicles. The paper also discusses the communication requirements of V2I systems, emphasizing the need for low-latency, reliable data exchange to ensure timely coordination between vehicles and signals. Technologies such as Dedicated Short-Range Communication and cellular V2X are mentioned as potential enablers, though integration with other low-power IoT technologies could further extend system scalability. The study highlights the importance of cybersecurity and data synchronization to maintain system reliability in real-world deployments. Overall, Xu et al. provide a strong theoretical and simulation-based foundation for cooperative traffic control in the context of intelligent transportation systems. Their work demonstrates how V2I cooperation can transform traditional intersections into smart, adaptive nodes that interact seamlessly with automated vehicles.

5. The paper “**Implementation and Performance Measurement of a V2X Communication System for Vehicle and Pedestrian Safety**” by Zishan Liu, Zhenyu Liu, Zhen Meng, Xinyang Yang, Lin Pu, and Lin Zhang presents a comprehensive study on the design, implementation, and evaluation of a V2X communication framework aimed at enhancing road safety for both vehicles and pedestrians. The research focuses on the integration of V2V and V2P communication using Dedicated Short-Range Communication and cellular-based V2X technologies. The authors propose V2X system architecture that allows real-time information exchange between vehicles, roadside infrastructure, and pedestrian devices such as smartphones. The goal is to prevent accidents by providing early collision warnings and cooperative awareness messages. A major contribution of this study is its focus on pedestrian safety, an area often under represented in V2X research.

By enabling communication between vehicles and mobile pedestrian devices, the system can issue warning alerts to both drivers and pedestrians in potential collision zones. The authors also highlight several challenges in deploying V2X systems, including network congestion, synchronization, interference, and standardization across multiple communication protocols. Security and privacy protection for user data are emphasized as crucial for large-scale implementation.

6. The paper “**LoRaWAN for Vehicular Networking: Field Tests for Vehicle-to-Roadside Communication**” by Gabriele Di Renzone, Stefano Parrino, Giacomo Peruzzi, Alessandro Pozzebon, and Lorenzo Vangelista investigates the potential of LoRaWAN technology to support vehicular communication, specifically Vehicle-to-Infrastructure links between moving vehicles and roadside gateways. The study aims to assess whether LoRaWAN known for its low power consumption, long-range connectivity, and low deployment cost can effectively contribute to Intelligent Transportation Systems . The authors conduct extensive field experiments using commercial LoRaWAN devices to evaluate communication performance in realistic vehicular environments. Parameters such as signal strength (RSSI), packet delivery ratio (PDR), latency, and data rate were measured across various speeds, distances, and environmental conditions. The field tests were carried out in both urban and suburban settings, allowing for comparison of LoRaWAN’s effectiveness under different propagation scenarios. Overall, this paper demonstrates that LoRaWAN is a promising complementary technology for V2I communication in ITS frameworks, particularly in low-data-rate, large-area, or rural environments. Their work provides valuable experimental evidence that supports the integration of LoRa-based networks into broader V2X ecosystems, helping to create cost-efficient and energy aware vehicular communication systems that enhance connectivity beyond the limits of traditional infrastructure.

7. The paper “**LoRa Architecture for V2X Communication: An Experimental Evaluation with Vehicles on the Move**” by Khandaker Foysal Haque, Ahmed Abdelgawad, Venkata Prasanth Yanambaka, and Kumar Yelamarthi explores the use of LoRa technology as a cost-effective and energy-efficient solution for V2X communication. The study focuses on evaluating how LoRa-based architectures perform when vehicles are in motion, addressing one of the major challenges in integrating LoRa into dynamic vehicular environments. The authors proposes LoRa-based V2X communication framework, where vehicles equipped with LoRa transceivers exchange information with each other and with roadside infrastructure.

The system is designed to support low-power, long-range, and low-data-rate communication, making it suitable for non-safety-critical vehicular applications such as traffic monitoring, vehicle diagnostics, and environmental data sharing. Overall, Haque et al. provide important experimental evidence supporting the feasibility of LoRa for vehicular networks. This research confirms that while LoRa cannot replace high-speed, low-latency systems required for safety applications, it is a promising candidate for supplementary V2X communication in smart transportation systems. The study contributes to the broader understanding of how LoRaWAN and LoRa-based architectures can enhance connectivity, scalability, and energy efficiency in next-generation vehicular communication frameworks.

8. The paper “**Implementation and Testing of V2I Communication Strategies for Emergency Vehicle Priority and Pedestrian Safety in Urban Environments**” by Federica Oliva, Enrico Landolfi, Alfredo Troiano, Giovanni Salzillo, Alfredo Massa, and Simone Mario D’Onghia focuses on the development and real-world testing of Vehicle-to-Infrastructure (V2I) communication systems designed to enhance emergency vehicle management and pedestrian safety in complex urban traffic conditions. The study emphasizes how real-time cooperation between vehicles and traffic infrastructure can significantly improve road safety, reduce response times for emergency vehicles, and minimize the risk of pedestrian accidents. The authors present a prototype V2I communication framework where emergency vehicles, traffic signals, and pedestrian crossing systems interact through wireless communication based on Dedicated Short-Range Communication. The system enables emergency vehicles to send priority requests to traffic lights along their route, allowing signals to switch in favor of their movement and create a “green wave” to facilitate faster passage. Simultaneously, the infrastructure communicates with roadside pedestrian units to ensure safe crossing control and warning alerts when emergency vehicles approach intersections. Overall, Oliva et al. deliver a significant contribution to intelligent transportation systems research by demonstrating how V2I communication can be effectively utilized for emergency vehicle prioritization and pedestrian protection. Their work bridges the gap between conceptual V2I models and deployable real-world solutions, advancing the development of smart urban mobility systems that combine automation, connectivity, and safety.

9. The paper “**Design, Implementation and Experimental Investigation of a Pedestrian Street Crossing Assistance System Based on Visible Light Communications**” by Alin-Mihai Căilean , Catalin Beguni, Sebastian-Andrei Avatamanit, Mihai Dimian and Valentin Popa propose a pedestrian street-crossing assistance system that uses Visible Light Communication to enhance road safety through infrastructure-based information exchange. In the context of Vehicle-to-Everything communication, this study demonstrates how existing LED traffic lights can act as communication nodes broadcasting safety messages to vehicles and pedestrians using modulated light. The authors highlight a key challenge in modern V2X systems: reliable communication in dense or interference-prone environments. VLC, which uses visible light instead of radio waves, offers high data security, low interference, and precise localization, making it suitable for short-range, line-of-sight communication scenarios such as pedestrian crossings. The proposed system integrates pedestrian detection with a VLC transmitter embedded in LED signs, which sends encoded safety messages to approaching vehicles equipped with VLC receivers. From a V2X perspective, the system serves as a localized safety application, where data is exchanged directly between infrastructure and nearby users. Its main strengths lie in cost efficiency and spatial accuracy. However, limitations such as line-of-sight dependence, sensitivity to weather and lighting conditions remains as open challenges. In conclusion this paper demonstrates a practical step toward integrating VLC into the broader V2X ecosystem. It shows how light-based communication can complement traditional RF-based systems to create smarter, safer, and more connected transportation environments particularly for pedestrian safety at urban crossings.

10. The paper “**LoRa-Based Traffic Flow Detection for Smart-Road**” by David Asiain and Diego Antolin propose traffic flow detection using LoRa technology as part of a smart-road infrastructure. Their work contributes to the growing field of V2X communication, where vehicles, infrastructure, and sensors collaborate to improve traffic management and road safety. The proposed system demonstrates how Low-Power Wide-Area Network technologies like LoRa can be effectively used for real-time monitoring of vehicle flow and road conditions key components of V2I communication. The authors highlight the limitations of traditional V2X communication methods such as Wi-Fi, Bluetooth, or cellular networks, which often suffer from high energy consumption, limited range, and higher operational costs. LoRa, on the other hand, provides long-range communication with minimal power usage, making it ideal for large-scale, continuous sensing across smart roads.

In conclusion, this research demonstrates that LoRa networks can serve as a low-cost, energy efficient backbone for smart-road and V2X applications. Their study proves that long-range IoT communication technologies can complement conventional vehicular networks, contributing to safer, more efficient, and data-driven transportation ecosystems.

11. The paper “**A LoRa-Based Mesh Network for Peer-to-Peer Long-Range Communication**” by Riccardo Berto, Paolo Napoletano, and Marco Savi introduce a LoRa-based mesh networking architecture designed to enable peer-to-peer (P2P) long-range communication without dependence on centralized gateways. This paper directly contributes to the development of V2V and V2X systems by addressing a critical challenge in vehicular networks maintaining reliable and scalable communication in areas lacking coverage, such as highways or remote roads. The proposed system uses LoRa transceivers operating in unlicensed ISM bands and introduces a routing protocol that allows nodes to forward packets through neighboring nodes until they reach their destination. This decentralized approach ensures continuous connectivity even when gateway links fail it’s an essential feature for V2V and V2X safety applications. In conclusion, the work demonstrates that LoRa-based mesh networking can effectively extend the reach and robustness of V2X systems. By enabling peer-to-peer, long-range, and low-power communication, it provides a practical foundation for cost-efficient, infrastructure-independent vehicular connectivity, supporting the vision of safer and smarter transportation networks.

12. The paper “**Multi-hop Communication between LoRa End Devices**” by Yosefine Triwidyastuti, Fifin Ernawati, and Charisma Dimas Affandi proposes multi-hop communication among LoRa end devices to overcome the range and coverage limitations of traditional star-topology LoRaWAN networks. This work shows how multi-hop routing can extend LoRa’s communication distance and reliability without relying only on gateways. The authors aim to enhance data transmission between distant nodes by allowing intermediate devices to act as relays, creating a flexible and adaptive communication path. In the V2X context, such a setup enables V2V or V2I data exchange even beyond direct line-of-sight, supporting applications like emergency alerts, rural traffic monitoring or cooperative awareness in areas lacking cellular coverage. The study emphasizes LoRa’s potential as a cost-effective complementary layer to existing RF-based V2X systems, providing extended connectivity and resilience for future smart-road and IoT-enabled transportation networks.

CHAPTER 3

DESIGN AND EXPLANATIONS

3.1 BLOCK DIAGRAM

3.1.1 CAR UNIT

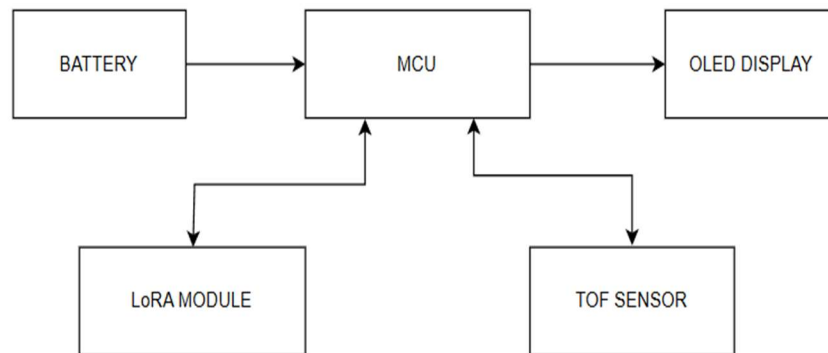


Fig 3.1.1 Car Unit

- **MCU (Microcontroller Unit)**

The **MCU** is the central processing unit of the car module. It collects data from all sensors, processes it, and coordinates communication with other vehicles or roadside units. It acts as the **brain** of your system interpreting sensor inputs (like motion or acceleration), displaying alerts, and transmitting messages via the LoRa module.

- **TOF Sensor**

A Time-of-Flight (ToF) sensor in a car unit is used to measure the distance between the vehicle and nearby objects using laser pulses. It emits an infrared light signal and calculates distance based on the time taken for the reflected light to return. In automotive systems, it helps in obstacle detection, collision avoidance, and parking assistance. Due to its fast response and high accuracy at short range, it improves vehicle safety and real-time decision-making.

- **LoRa Module**

The LoRa module enables wireless long-range, low-power communication between vehicles and infrastructure — forming the basis of V2X communication.

1. Vehicle-to-Vehicle (V2V): Alerts nearby vehicles about speed, position, or emergency events.
2. Vehicle-to-Infrastructure (V2I): Communicates with traffic lights or RSUs to receive signal information (e.g., red light ahead, ambulance priority).
3. Vehicle-to-Pedestrian (V2P): Sends alerts when pedestrians are detected.

- **OLED Display**

The OLED display provides a visual interface for the driver. It shows real-time alerts, such as:

1. Nearby vehicle warning (collision alert)
2. Pedestrian crossing detected
3. Traffic signal information from RSU
4. Ambulance or emergency vehicle alert

- **Battery**

The battery powers the entire car unit. It provides stable DC power to the MCU and peripheral components, ensuring reliable operation even when integrated with the vehicle's electrical system.

3.1.2 TRAFFIC UNIT

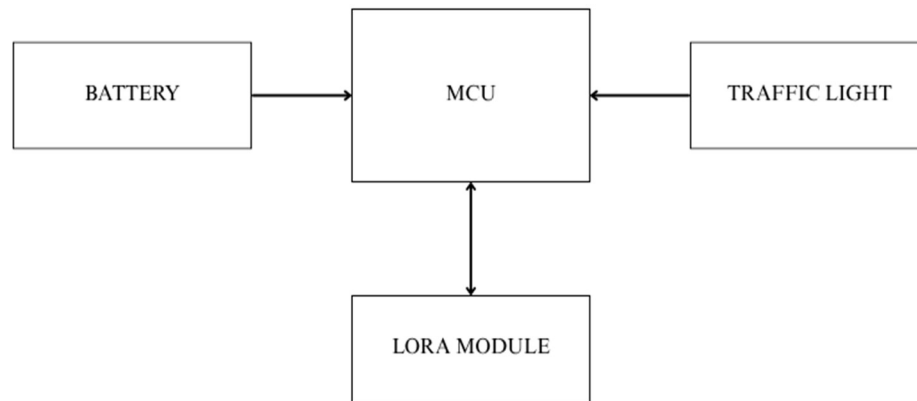


Fig 3.1.2 Traffic Unit

- **MCU (Microcontroller Unit)**

The MCU is the control and processing core of the traffic light unit. It manages the traffic light's operation and handles communication with vehicles through the LoRa module. The MCU reads the current traffic signal state (Red, Yellow, or Green) and converts it into digital data that can be transmitted wirelessly. It ensures that all vehicles approaching the intersection receive accurate, real-time signal information.

- **Traffic Light**

The traffic light is both a controlled device and a data source for the MCU.

1. The MCU reads the light's status (e.g., Red = Stop, Green = Go, Yellow = Prepare).
2. This information is packaged and sent via the LoRa module to nearby vehicles. This helps vehicles receive early alerts about signal changes improving safety and reducing the risk of red-light violations.

- **Battery**

The battery powers the entire traffic light unit, providing a stable DC supply to the MCU and LoRa module. It ensures uninterrupted operation even during power fluctuations or outages, making the system reliable in outdoor conditions.

- **LoRa Module**

The LoRa module enables long-range wireless communication between the traffic light unit and nearby vehicles or pedestrian units.

1. It transmits the traffic signal state to vehicles (V2I: Vehicle-to-Infrastructure).
2. It can also receive messages, such as emergency vehicle or ambulance priority requests, allowing the traffic controller to switch signals accordingly. LoRa's low power and long range make it ideal for real-time communication at intersections and smart-road environments.

3.1.3 RSU

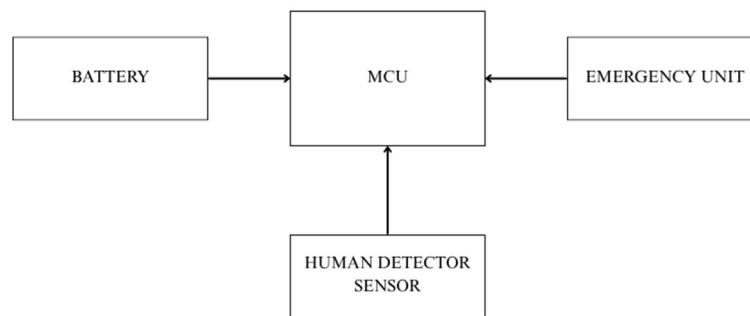


Fig 3.1.3 RSU

- **MCU (Microcontroller Unit)**

The MCU is the main controller and processing hub of the RSU. It receives input signals from the Human Detector Sensor to identify when a pedestrian is present at a crossing. It communicates with the Emergency Unit or nearby vehicles using wireless communication. The MCU ensures real-time decision-making, such as triggering alerts to approaching vehicles or changing traffic light signals to red when pedestrians are detected.

- **Battery**

The Battery supplies stable power to the MCU, sensors, and communication modules. Since RSUs are often placed roadside, battery or solar-powered operation ensures independent functioning even in case of grid power failure.

- **Human Detector Sensor**

This sensor is responsible for detecting the presence or movement of pedestrians near or on the crosswalk.

1. The ToF sensor emits infrared laser light and measures the time taken for the reflected signal to return. When a person enters its range, the reflected signal is detected, indicating pedestrian presence.
2. When a pedestrian is detected, the sensor sends a signal to the MCU. This data helps prevent accidents at pedestrian crossings and improves situational awareness for vehicles.

- **Emergency Unit**

The Emergency Unit handles critical alerts and priority communications.

1. It can send priority signals to the traffic control system for ambulance or emergency vehicle clearance.
2. It can also trigger warning signals to nearby vehicles via LoRa communication when pedestrians or emergency situations are detected. This makes it a vital component for V2P and V2I interactions.

3.2 CIRCUIT DIAGRAM

3.2.1 CAR UNIT

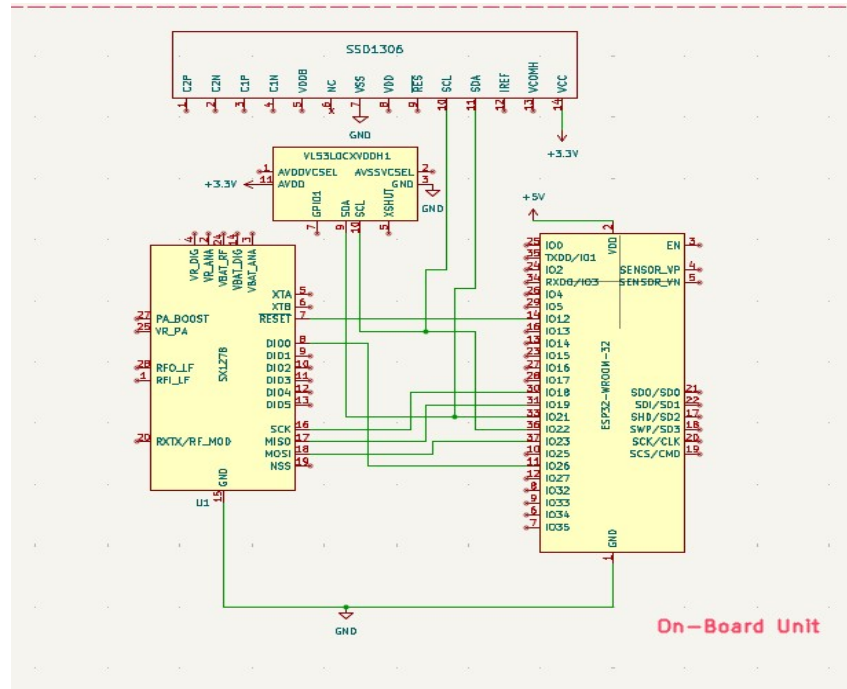


Fig 3.2.1 Car Unit

3.2.2 TRAFFIC UNIT

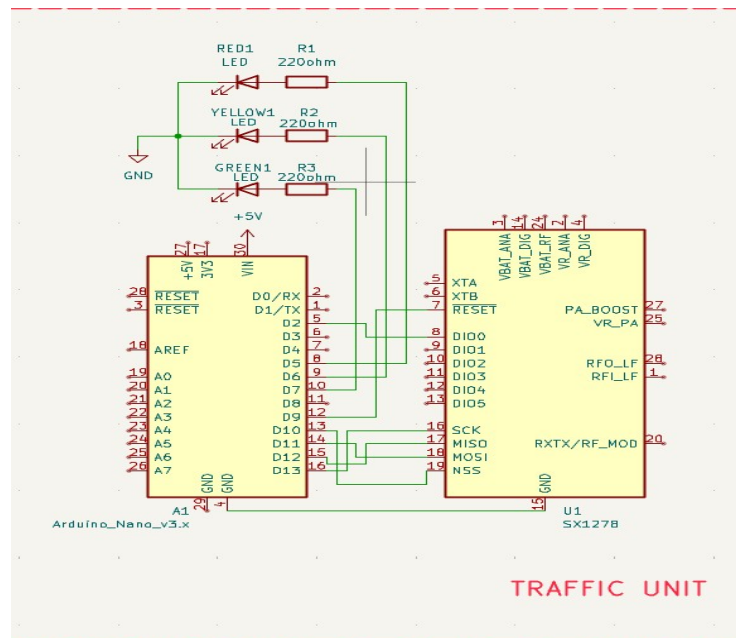


Fig 3.2.1 Traffic Unit

3.2.3 RSU UNIT

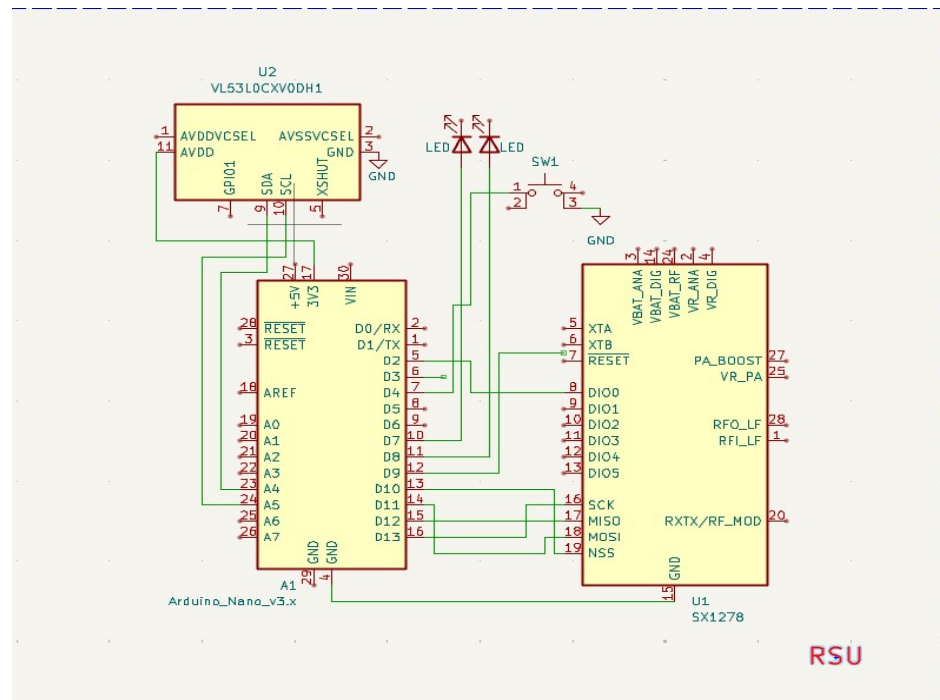


Fig 3.2.3 RSU

3.3 COMPONENTS USED

1. Arduino Nano:

The Arduino Nano is a compact, versatile, and cost-effective microcontroller board based on the ATmega328P microchip. It offers the same functionality as the Arduino Uno but in a much smaller form factor, making it ideal for embedded and space-constrained applications like vehicle and roadside communication systems. The board features 14 digital input/output pins, 8 analog inputs, a 16 MHz crystal oscillator, and supports communication protocols such as UART, SPI, and I2C. It operates on 5V, with a recommended input voltage range of 7–12V, and can be powered through a USB connection or an external battery. In this project, the Arduino Nano serves as the main controller for the vehicle. It processes sensor data and handles communication through the LoRa module.



Fig 3.3.1 Arduino Nano (amazon.in)

2. LoRa Module SX1278:

The SX1278 LoRa module is a long-range, low-power wireless transceiver based on SX1278 chip, designed for LoRa communication technology. It operates in the 433 MHz ISM band and uses spread spectrum modulation, which provides exceptional sensitivity and interference resistance, enabling reliable communication over several kilometers with minimal power consumption. The module supports both point-to-point and mesh networking modes, making it ideal for Vehicle to Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication in your project. It interfaces easily with microcontrollers like Arduino Nano or ESP32 through SPI communication. In this project, the SX1278 module wireless data transfer between the car unit, traffic light unit, and RSU, facilitating real-time alerts such as vehicle proximity warnings, traffic signal updates, and emergency notifications. Its long-range capability, low power usage, and robust signal performance make it a key component for building efficient and scalable smart-road communication networks.

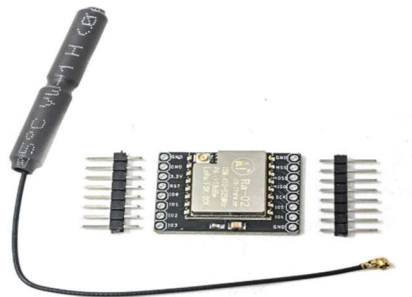


Fig 3.3.2 LoRa Module SX1278 (Robu.in)

3. VL53L0X Sensor:

The VL53L0X Time-of-Flight (ToF) Laser Distance Sensor is a compact and accurate distance measurement device commonly used in embedded systems and smart automation projects. It measures distance using laser-based time-of-flight technology, which calculates how long it takes for a pulse of infrared light to travel to an object and return to the sensor. Since the speed of light is known, the sensor determines the distance by measuring this tiny time interval. Unlike ultrasonic sensors that rely on sound waves, the VL53L0X uses a 940 nm invisible infrared laser, allowing it to provide faster and more precise short-range measurements. It typically measures distances from about 30 mm up to 2 meters, though reliable performance is usually achieved within 1 to 1.2 meters depending on lighting conditions and surface reflectivity. The sensor communicates through the I2C protocol, making it easy to interface with microcontrollers such as ESP32 and Arduino. One of its main advantages is its small size and high accuracy at close range. It is less affected by object color compared to traditional IR sensors and offers quicker response times than ultrasonic modules. However, its performance can decrease in bright sunlight or when detecting very dark surfaces. Overall, the VL53L0X is widely used in robotics, obstacle detection, smart vehicles, and IoT applications due to its precision, compact design, and ease of integration.

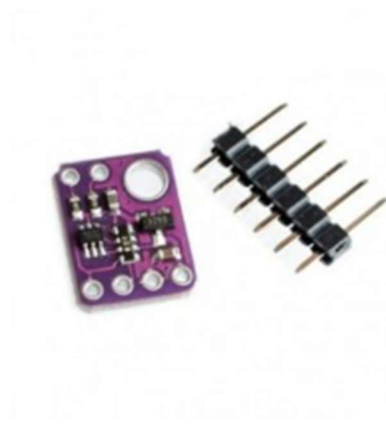


Fig 3.3.3 VL53L0X Sensor (robu.in)

4. LED (Light Emitting Diode):

LED is widely used as an indicator in electronic circuits due to its low power consumption, long life, and fast response time. In this project, LEDs are used in the traffic light unit to represent the signal states—Red (Stop), Yellow (Ready), and Green (Go). They help visualize the current traffic condition and also serve as visual indicators for communication or alert signals received through the LoRa module.

5. Switch:

A switch is an electromechanical device used to manually open or close an electrical circuit, thereby controlling the flow of current. In this project, the switch serves as a control or reset input for various functions such as powering the unit, activating the system, or simulating emergency scenarios (like ambulance alerts). It provides a simple and effective means for user control or testing during operation.

6. ESP WROOM 32

ESP32 is a series of low-cost, low-power system-on-chip microcontrollers with integrated Wi-Fi and dual-mode Bluetooth. The ESP32 series employs either a Tensilica Xtensa LX6 microprocessor in both dual-core and single-core variations, an Xtensa LX7 dual-core microprocessor, or a single-core RISC-V microprocessor and includes built-in antenna switches, RF balun, power amplifier, low-noise receive amplifier, filters, and power-management modules. It is commonly found either on device-specific PCBs or on a range of development boards with GPIO pins and various connectors depending on the model and manufacturer of the board. It is a successor to the ESP8266 microcontroller.

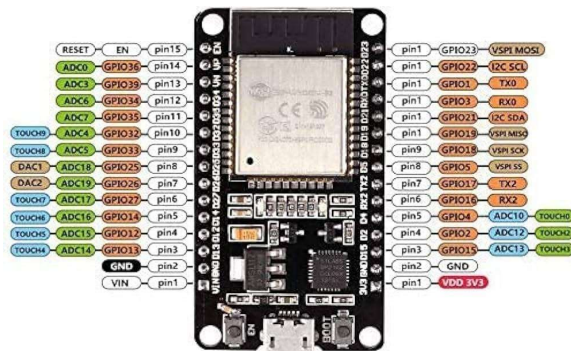


Fig 3.3.4 ESP32 S (circuits4you)

3.3.1 WORKING

- **CAR UNIT**

When two vehicles approach each other within a short distance, the ultrasonic sensor plays a crucial role in detecting their proximity. It continuously measures the distance between the host vehicle and any nearby object or vehicle by emitting ultrasonic waves and calculating the time taken for the reflected signal to return. Once the measured distance falls below a predefined safety threshold, the sensor sends this data to the Arduino microcontroller (MCU) for processing. The Arduino then analyzes the distance information and recognizes a potential collision risk or unsafe proximity. After identifying the risk, the Arduino communicates this event through the LoRa module. The LoRa module transmits an alert signal to nearby vehicles equipped with similar V2V units and to the Roadside Unit (RSU) for further broadcasting or logging. This real-time communication helps notify nearby vehicles about the detected threat, allowing them to take preventive actions, such as slowing down or changing lanes. This setup forms an essential part of V2X communication, enhancing road safety by enabling vehicles to share situational awareness. It ensures timely collision avoidance, driver alerting, proximity warnings, even in areas without cellular or internet coverage, contributing to a smarter and safer transportation network.

- **TRAFFIC LIGHT UNIT**

In the traffic light unit, the Arduino microcontroller (MCU) serves as the central controller that continuously monitors the current status of the traffic signal whether it is red, yellow, or green. Each light is connected to the Arduino through separate digital output pins that control the corresponding LED indicators. To enable long-range wireless communication, the LoRa (SX1278) module is interfaced with the Arduino. Whenever the traffic light changes its state, the Arduino sends a corresponding status message—for example, “Red Light Ahead,” “Yellow Light – Prepare to Stop,” or “Green Light – Proceed”—through the LoRa module. Vehicles equipped with LoRa receivers can interpret this message and display it on their onboard display or alert system, allowing drivers to be aware of the signal even before the traffic light becomes visible. Through this process, the system enhances Vehicle-to-Infrastructure communication by providing real-time traffic signal updates, reducing the risk of signal violations, and contributing to smoother and safer traffic flow by intelligent transportation systems.

- **RSU (ROAD SIDE UNIT)**

The Road Side Unit (RSU) plays a vital role as the central communication bridge and intelligent control hub in your V2X communication system. It serves as an intermediary between vehicles, pedestrians, and traffic infrastructure, ensuring seamless data flow and real-time decision-making for safer and smarter road management. The RSU is equipped with a microcontroller and a LoRa module that enable it to both receive and transmit wireless data across a long range. It continuously listens for incoming signals from vehicles that may include collision warnings, vehicle proximity alerts, or emergency vehicle notifications. When the RSU receives such data, it processes the information and decides the appropriate response based on situation. By managing these multiple communication flows V2V, V2I and V2P, the RSU acts as the decision-making and synchronization center of the network. It enhances the overall reliability of the V2X system by ensuring that all connected units operate cohesively, enabling real-time awareness, reducing accidents, and supporting the foundation for future smart transportation and intelligent traffic control systems.

3.4 FLOW CHART

3.4.1 RSU (ROAD SIDE UNIT)

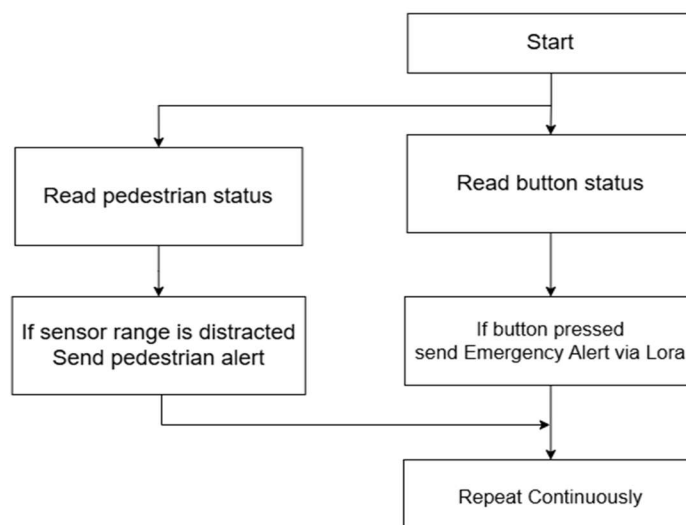


Fig 3.4.1 Flowchart of RSU

3.4.2 TRAFFIC SIGNAL UNIT

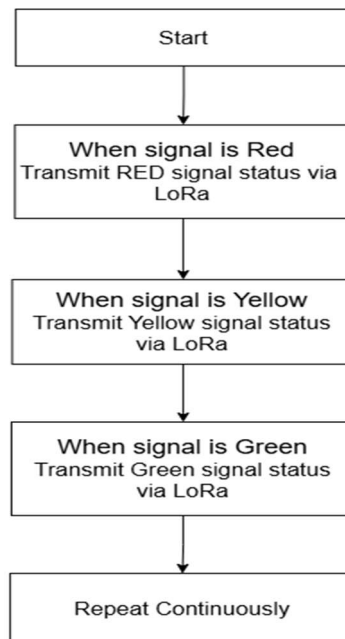


Fig 3.4.2 Flowchart of Traffic signal unit

3.4.3 CAR UNIT

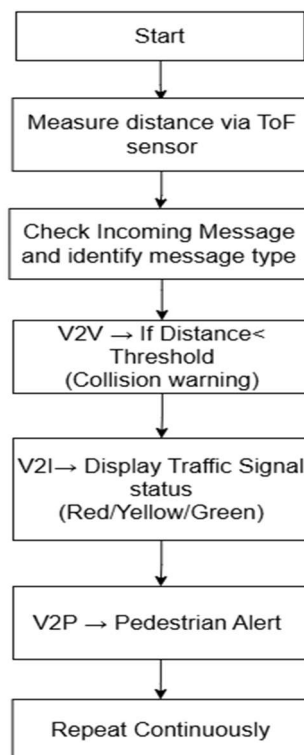


Fig 3.4.3 Flowchart of car unit

CHAPTER 4

RESULT

The developed LoRa-based Intelligent Transportation System was successfully implemented and tested for Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), and Vehicle-to-Pedestrian (V2P) communication. The system demonstrated reliable wireless communication between different units using SX1278 LoRa modules.

In the V2V module, distance information between vehicles was transmitted and received effectively. When the distance between vehicles fell below the predefined threshold, the system successfully generated a collision alert, which was displayed on the OLED screen along with an audible buzzer indication. This confirms the system's ability to provide real-time safety warnings to prevent potential accidents.

In the V2I communication module, the traffic unit transmitted real-time signal status (Red, Yellow, Green) using LoRa. The car unit received this information accurately and displayed appropriate messages such as "STOP," "READY," and "GO." This demonstrates the effectiveness of infrastructure-assisted communication in improving driver awareness and reducing signal violations.

In the V2P module, the RSU detected pedestrian presence using the human detection sensor and transmitted alerts to nearby vehicles. The car unit successfully received these alerts and generated visual and audio warnings, ensuring enhanced pedestrian safety. Additionally, the emergency alert feature allowed manual triggering of an ambulance warning, which was correctly received and displayed by the vehicle unit. The communication range of the LoRa modules was found to be sufficient for short to medium distances in the experimental setup, with stable data transmission and minimal packet loss. The system operated with low power consumption, making it suitable for practical deployment in real-world traffic environments.

However, due to the inherent limitations of LoRa technology, such as low data rate and higher latency compared to conventional communication systems, the system is best suited for low-data-rate safety applications rather than high-speed real-time control. Overall the proposed system successfully demonstrates a cost-effective and efficient LoRA based V2X communication framework that enhances road safety, improve traffic awareness, and support smart transportation applications.

4.1 CAR UNIT



Fig 4.1 Car unit

4.2 TRAFFIC UNIT

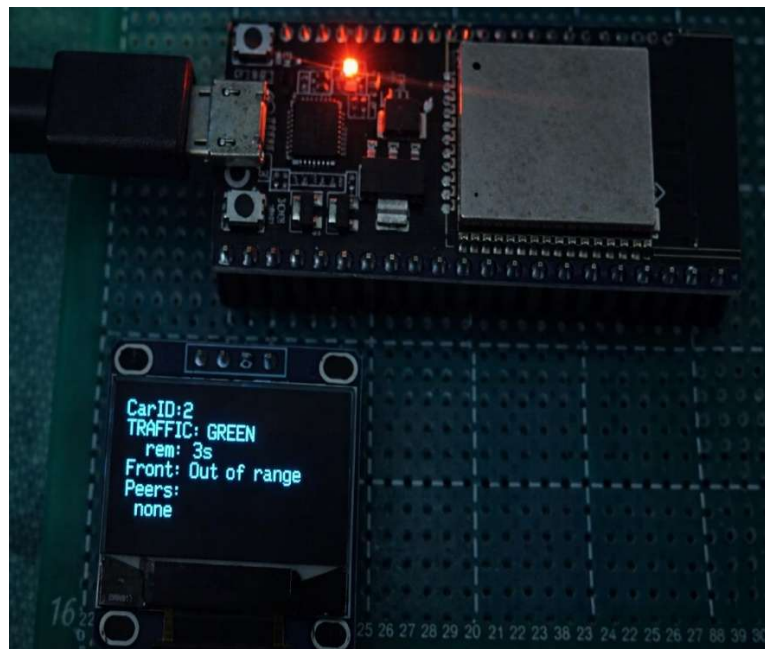
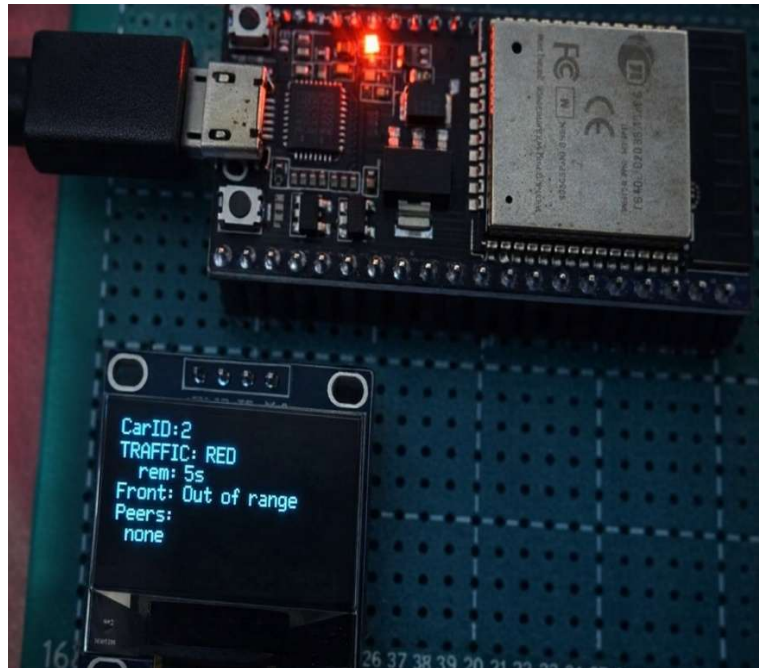


Fig 4.2 Traffic unit

4.3 RSU

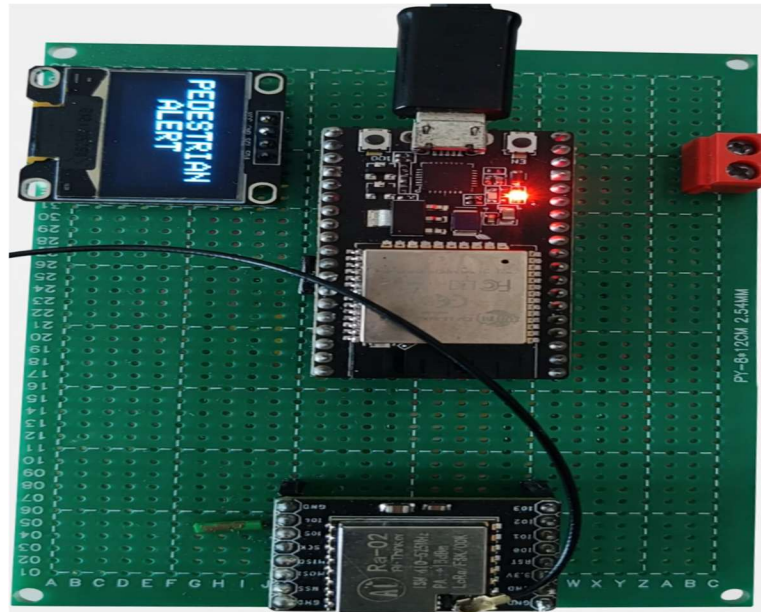


Fig 4.3.1 Pedestrian alert

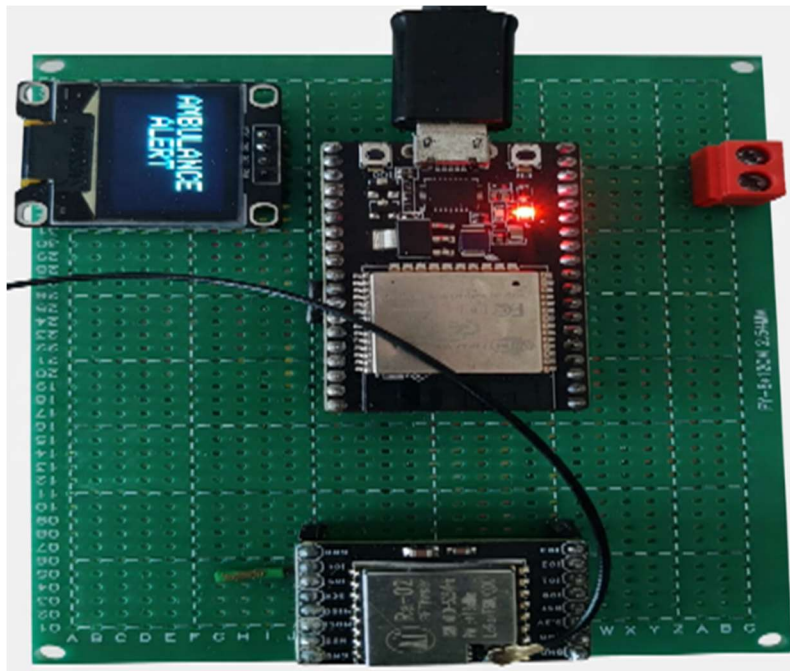


Fig 4.3.2 Ambulance alert

CHAPTER 5

CONCLUSION

The proposed V2X (Vehicle-to-Everything) communication system using LoRa technology has proven to be an efficient, low-cost, and reliable solution for enhancing road safety and traffic management. By enabling real-time communication between vehicles, traffic lights, and roadside units (RSU), the system effectively supports Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), and Vehicle-to-Pedestrian (V2P) communication. Through experimental implementation, the project demonstrated the ability to detect nearby vehicles using ultrasonic sensors, transmit traffic signal information, and provide pedestrian and emergency alerts with the help of LoRa-based wireless communication.

The system successfully minimizes latency, operates efficiently over long distances, and functions reliably even in areas without cellular coverage. This project highlights the potential of LoRa as a communication backbone for intelligent transportation systems (ITS). With further development, such as integrating GPS, cloud data logging, and advanced sensor fusion, the system can evolve into a scalable solution for future smart city and autonomous vehicle applications.

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